

Bayesian Inference of Mixing and Transmission Heterogeneity in Stratified Disease Surveillance Models

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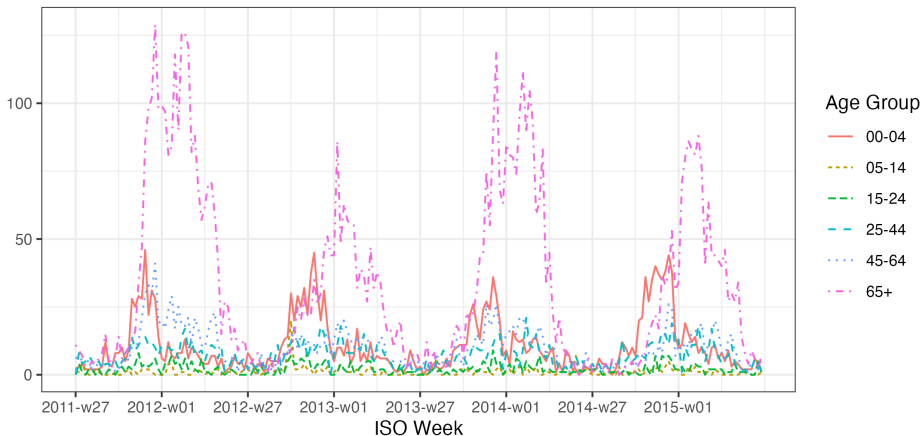
<https://arxiv.org/abs/2605.29081>

Stratified Surveillance Data

- Disease surveillance increasingly records incident counts by both geography *and* additional categorical demographics (e.g. age group).
- These “**doubly-stratified**” panels are scientifically valuable:
 - predict disparities in health outcomes, by subpopulation
 - learn who acquires infection from whom (“WAIFW”)
 - detect vulnerable groups early in “real time”
- ...and statistically challenging:
 - no true replication, each cell contributes one count per week
 - fine partitioning may lead to zero-inflation, esp. in rare disease settings
 - fine partitioning threatens large-population approximations
- How do we borrow information to learn subpopulation-level transmission mechanism *without* imposing rigid mixing assumptions?

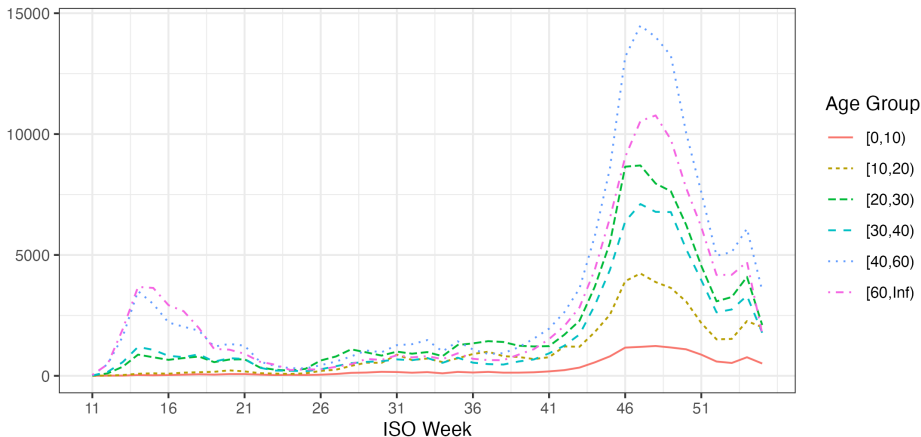
“Rare-Disease” Incidence Paradigm

Reported Incidence of Norovirus in Berlin, July 2011 to July 2015



“Outbreak-Driven” Incidence Paradigm

Reported Incidence of COVID-19 in MI Lower Peninsula, March 2020 to Jan 2021



Discrete-Time Incidence Models

- Why not use existing models?
 - TSIR, Renewal-theory formulations [FG00; BFG02; Bjø+19]
 - Endemic-Epidemic (hhh4) [KR03; HHH05; HP12; MHH17; ADS26]
 - Binomial Chains [DG99; Jew+23]
- (a) Application is univariate or stratified *only* by geog.
 - Mixing among *categories* is unstructured, unlike geog. mixing
 - Large-sample approximations are less suitable for doubly-stratified data
- (b) Transmission mechanism lacks robust treatment of subpopulation co-occurrence (e.g. a collection of independent processes with unique renewal equations)
- (c) Analytical intractability, esp. in binomial chain formulations
- (d) Heavy-tailed offspring dist. may point to individual-level heterogeneity in transmission (“**transmission heterogeneity**”) [Llo+05; Zel+20; ZBZ22; CSD25]

Modeling Contribution

- Use Y_{tgi}^* to denote **incidence** at time t for age group i in geographic unit g
 - Use Y_{tgi} to denote **prevalence** and X_{tgi} the susceptible pool size
- (1) Transmission heterogeneity (“super-spreading aware”) Linear Predictor:
improve on the EE model formulation
- $$\mathbb{E}\left(Y_{tgi}^* | \mathbf{Y}_{(t-1)}^*\right) = \delta_{tgi} + \varphi_{tgi} \sum_{g', i'} w_{gg'}^{(G)} w_{ii'}^{(I)} Y_{(t-1)g'i'}^*$$
- (2) Learn age-group mixing weights $w_{ii'}^{(I)}$ without a strict low-dimensional parameterization or additional data streams
- (3) Define a model class broad enough for these components to be useful in both a rare-disease *and* outbreak-driven paradigm
- Explicit adjustment of mean incidence based on susceptible pool size X_{tgi} or population size $E_{gi} > X_{tgi}$

Latent-Variable Transmission Mechanism

- Use Y_{tgi}^* to denote **incidence** at time t for age group i in geographic unit g
- Use Y_{tgi} to denote **prevalence** and X_{tgi} the susceptible pool size
- Y_{tgi}^* is now driven by a latent “infectiousness” contribution r_{tgi}
- Individual infectious potentials aggregate to a Gamma cell-level variable:

$$(r_{tgi} \mid Y_{tgi}) \stackrel{\text{indep}}{\sim} \text{Gamma} \left(\frac{R_{tgi} Y_{tgi}}{\theta}, \theta \right).$$

- The infectious load sent into recipient cell (g, i) is

$$\lambda_{tgi}(\mathbf{r}_t) = \delta_{tgi} + \varphi_{tgi} \sum_{g', i'} w_{gg'}^{(G)} w_{ii'}^{(I)} r_{tg' i'}.$$

- Mean incidence follows a disease-specific “link” func.:

$$\mathbb{E}(Y_{tgi}^* \mid X_{tgi}, \mathbf{r}_t) = g(X_{tgi}, \lambda_{tgi}(\mathbf{r}_t)).$$

- Will need to replace Y_{tgi} and X_{tgi} with estimates obtained from data

EE Model Limiting Case

- In rare-disease settings, assume negligible susceptible depletion and one-period infectiousness:

$$\hat{X}_{tgi} = E_{gi}, \quad \hat{Y}_{tgi} = Y_{(t-1)gi}^*.$$

- As $\theta \downarrow 0$, latent infectiousness collapses to a deterministic function of lag-1 incidence.
- With an identity “link” and negative-binomial likelihood, the conditional mean becomes the familiar doubly-stratified EE mean:

$$\mathbb{E}(Y_{tgi}^* \mid \mathbf{Y}_{t-1}^*) = \delta_{tgi} + R_0 \varphi_{tgi} \sum_{g', i'} w_{gg'}^{(G)} w_{ii'}^{(I)} Y_{(t-1)g'i'}^*.$$

- The proposed model induces additional marginal covariance without changing mean structure.

Transmission Heterogeneity Moments

- Marginalizing over $\mathbf{r}_t$ leaves the rare-disease mean unchanged but increases marginal variance:

$$\text{Var}(Y_{tgi}^* \mid \mathbf{Y}_{t-1}^*, \theta > 0) = \text{Var}(Y_{tgi}^* \mid \mathbf{Y}_{t-1}^*, \theta = 0) + \theta \mathbb{V} + \theta \psi \mathbb{V}.$$

- It also induces positive covariance between recipient cells that share upstream sources:

$$\text{Cov}(Y_{tgi}^*, Y_{tab}^* \mid \mathbf{Y}_{t-1}^*) = \theta R_0 \varphi_{tgi} \varphi_{tab} \sum_{g', i'} w_{gg'}^{(G)} w_{ii'}^{(I)} w_{ag'}^{(G)} w_{bi'}^{(I)} Y_{(t-1)g'i'}^*.$$

- This is the feature classical EE models lack: shared latent infectiousness creates clustered subpopulation behavior.

Generative Age-Mixing Prior

- The estimand is the between-age mixing matrix $\mathbf{w}^{(I)}$.
- The model samples an unobserved symmetric, elementwise-positive contact-rate matrix $\mathbf{C}$.
- Mixing weights are obtained by column-normalization:

$$w_{ii'}^{(I)} \propto C_{ii'}.$$

- Gamma priors on entries of $\mathbf{C}$ induce Dirichlet priors on columns of $\mathbf{w}^{(I)}$.
- The prior can regularize toward assortative, homogeneous, or externally-informed mixing without constraining the posterior to a one-dimensional deformation of a known contact matrix.

Rare-Disease and Outbreak Specifications

	Rare-disease specification	Outbreak-driven specification
Use case	Endemic, low incidence per capita	Epidemic waves and finite susceptible pools
Example	Berlin norovirus, 2011–2015	Michigan COVID-19, Mar. 2020–Jan. 2021
Population estimates	$\hat{X}_{tgi} = E_{gi}$ $\hat{Y}_{tgi} = Y_{(t-1)gi}^*$	$\hat{X}_{tgi} = E_{gi} - \sum_{d < t} Y_{(t-d)gi}^*$ $\hat{Y}_{tgi} = \sum_{d < t} e^{-\gamma(d-1)} Y_{(t-d)gi}^*$
Likelihood/link for $Y_{tgi}^* \mid \mathbf{r}_t$	$\text{NegBin}(\lambda_{tgi}, \psi)$	$\text{BetaBinom}(\hat{X}_{tgi}, 1 - e^{-\lambda_{tgi}}, k)$
Primary purpose	Compare against doubly-stratified EE	Preserve finite-population support and learn mixing without external contact data

Simulation Study Design

- Data mimic the Berlin norovirus panel:

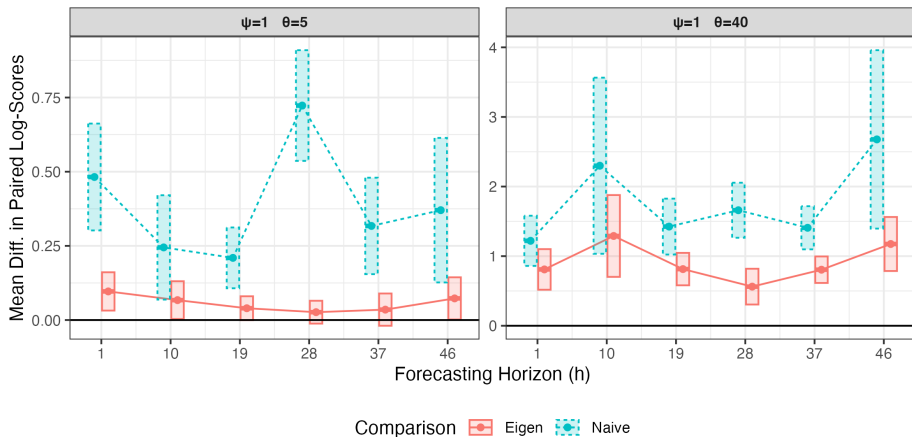
$$G = 12, \quad I = 6, \quad T = 208, \quad H = 52.$$

- Sixteen scenarios vary latent and observation-level dispersion:

$$(\theta, \psi) \in \{0.05, 5, 15, 40\} \times \{0.05, 0.5, 1, 3\}.$$

- Each scenario uses 200 independent datasets.
- Three models are compared:
 - **Full:** latent infectiousness and generative $\mathbf{w}^{(I)}$
 - **Reduced:** Meyer-Held EE comparator
 - **Naive:** endemic-only baseline
- Forecasting is evaluated by multivariate logarithmic score [GR07].

Forecasting Performance



Positive paired log-score differences favor the full model over the naive or reduced comparators. The advantage grows with latent dispersion θ , especially when observation-level dispersion ψ is small.

Simulation Coverage Diagnostics

- Observation-level dispersion ψ is covered near the nominal 90% level by the full model across most scenarios.
- The reduced model increasingly undercovers ψ as latent heterogeneity grows because it must absorb missing latent variation elsewhere.
- Latent dispersion θ is hardest when the truth is near the boundary or extremely large relative to ψ .
- The new layer helps most when the data-generating process contains real transmission heterogeneity; boundary cases mainly reveal prior and computation sensitivity.

Berlin Norovirus Application

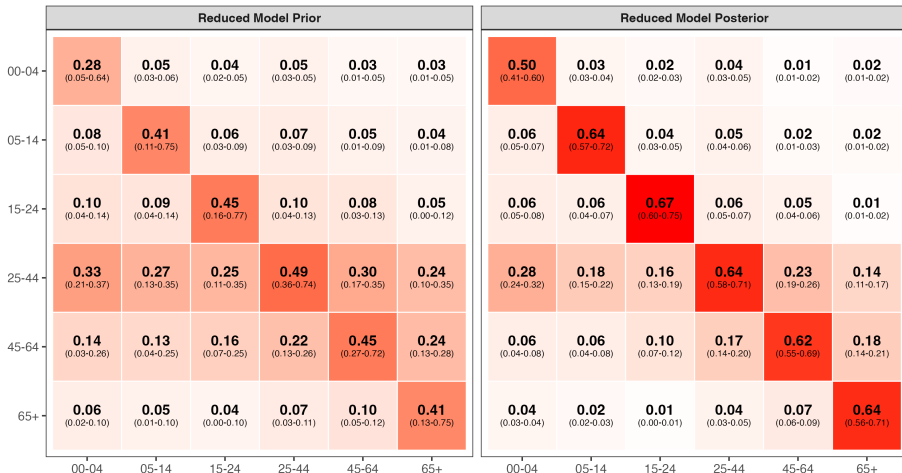
- Weekly norovirus gastroenteritis counts in Berlin, 2011–2015.
 - Dataset and model obtained from [MH17]
 - $T = 208$ weeks, $G = 12$ administrative districts, and $I = 6$ age groups
- * Why use the Meyer-Held model as a comparator?
 - Very few published models are close enough for comparison
 - The MH model assumes $\mathbf{w}^{(I)}$ is a one-dimensional deformation of an existing contact matrix (see POLYMOD Contact Survey, [Mos+08])
- The full model is compared to a Bayesian version of Meyer-Held and the original `hhh4contacts` fit.
 - Stan fits used $16 \text{ chains} \times (2,000 \text{ warmup} + 2,000 \text{ retained draws})$
 - Diagnostics passed routine inspection with $\max(\hat{R}) < 1.01$.

Norovirus Parameter Inference

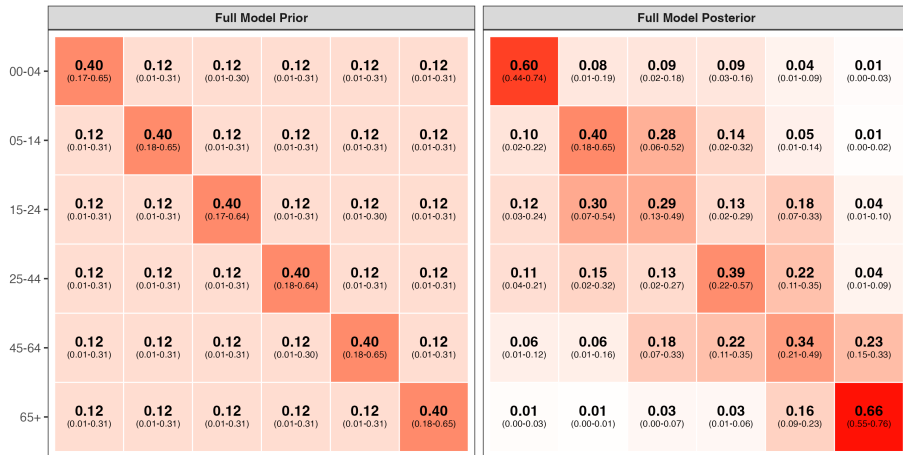
Parameter	Full model	Meyer-Held Bayes	hhh4contacts
$w_{11}^{(I)}$	0.60 (0.44–0.74)	0.50 (0.41–0.60)	0.49 (0.40–0.60)
$w_{22}^{(I)}$	0.40 (0.18–0.65)	0.64 (0.57–0.72)	0.64 (0.56–0.73)
$w_{12}^{(I)}$	0.07 (0.01–0.19)	0.04 (0.03–0.04)	0.04 (0.03–0.04)
ψ	0.19 (0.16–0.22)	0.31 (0.29–0.34)	0.31 (0.29–0.32)
θ	6.89 (5.14–9.27)	–	–

- The full model reproduces strong assortative age mixing while allowing wider, higher-dimensional uncertainty.
- The posterior for θ concentrates away from the EE limit, so extra variation is partitioned into latent transmission heterogeneity and observation noise.

Norovirus Age-Mixing Posterior (1/3)

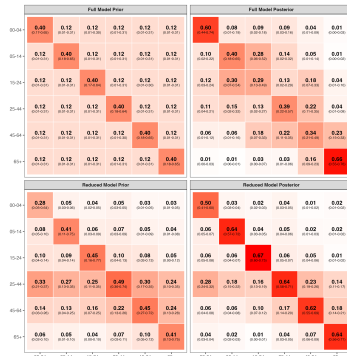


Norovirus Age-Mixing Posterior (2/3)

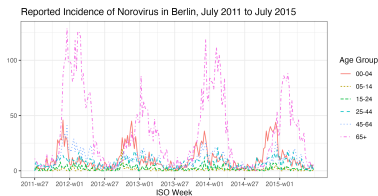


Norovirus Age-Mixing Posterior (3/3)

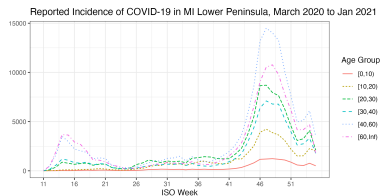
- Diagonal entries remain large, consistent with age-assortative transmission.
- Posterior intervals are wider because $w^{(I)}$ is learned from incidence data rather than confined to a one-dimensional contact-matrix deformation.
- The pattern is consistent with earlier evidence that norovirus dynamics can reflect source-and-sink heterogeneity.



Outbreak-Driven COVID-19 Application



Berlin norovirus: persistent endemic structure.



Michigan COVID-19: epidemic waves and policy-era shifts.

- Michigan counts are sufficiently high in some cells that the bound $Y_{tgi}^* \leq X_{tgi}$ matters.
- No population-specific external age-contact matrix is available, so age mixing must be learned directly.

Bounded Outbreak Specification

- Data are weekly COVID-19 cases in Michigan, March 2020–January 2021:

$$T = 45, \quad G = 66 \text{ PUMAs}, \quad I = 6 \text{ age groups.}$$

- Susceptible depletion is permanent over the analysis window:

$$\hat{X}_{tgi} = E_{gi} - \sum_{d=1}^{t-1} Y_{(t-d)gi}^*.$$

- Infectious prevalence is estimated by a distributed lag:

$$\hat{Y}_{tgi} = \sum_{d=1}^{t-1} e^{-\gamma(d-1)} Y_{(t-d)gi}^*.$$

- Incidence is beta-binomial with a finite-population support and extra-binomial precision k :

$$Y_{tgi}^* \mid \hat{X}_{tgi}, \mathbf{r}_t \stackrel{\text{indep}}{\sim} \text{BetaBinom} \left(\hat{X}_{tgi}, p_{tgi}, k \right).$$

Mixing and Infection Probability

- Given incoming infectiousness, the infection probability is

$$p_{tgi} = 1 - \exp \left(- \left(\delta_{gi} + \frac{1}{E_{gi}} \sum_{g', i'} w_{gg'}^{(G)} w_{ii'}^{(I)} r_{tg' i'} \right) \right).$$

- Between-age mixing again comes from the generative contact matrix prior for $\mathbf{C}$.
- Age-specific activity α_i is tied to column totals of $\mathbf{C}$, giving the data another route to learn contact structure.
- Geographic mixing uses destination attractiveness and source-specific distance decay:

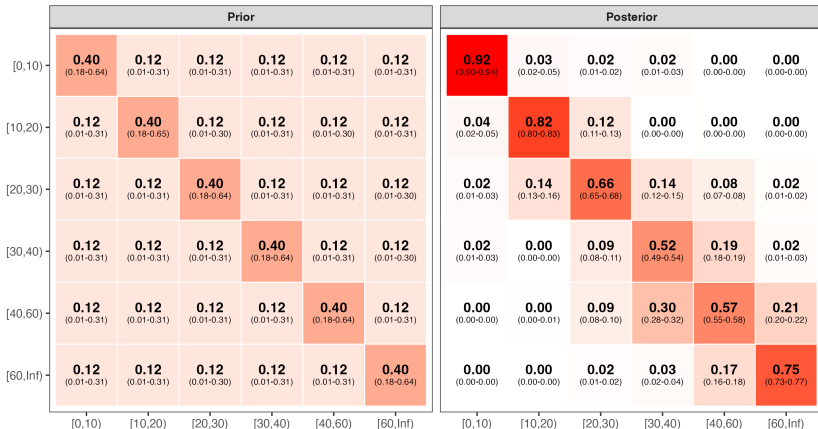
$$w_{gg'}^{(G)} \propto \tau_g D_{gg'}^{-\rho_{g'}}.$$

COVID-19 Parameter Inference

Parameter	Prior	Posterior
$w_{11}^{(I)}$	0.40 (0.18–0.64)	0.92 (0.90–0.94)
$w_{22}^{(I)}$	0.40 (0.18–0.65)	0.82 (0.80–0.83)
$w_{12}^{(I)}$	0.12 (0.01–0.31)	0.03 (0.02–0.05)
γ	0.99 (0.08–13.07)	1.64 (1.49–1.81)
θ	5.07 (0.40–62.38)	15.67 (14.54–16.89)
$k/10^4$	1.00 (0.08–12.60)	7.02 (6.24–7.98)

- The posterior for θ is far from the EE limit, indicating substantial latent transmission heterogeneity.
- The beta-binomial precision still allows extra-binomial variation, but low $\text{Cor}(\theta, k) \approx 0.01$ suggests the two dispersion mechanisms are not redundant.

COVID-19 Age-Mixing Posterior



Posterior age-mixing weights concentrate much more sharply on within-age transmission than the mildly assortative prior.

COVID-19 Mechanistic Interpretation

- Strong diagonal $w^{(I)}$ estimates indicate age-assortative mixing learned from incidence alone.
- γ has a 90% credible interval of (1.49, 1.81), so estimated infectious prevalence is dominated by lag-1 and lag-2 incidence.
- The fitted process explains epidemic dynamics primarily through structured feedback and latent infectiousness, not only observation-level noise.
- The outbreak specification therefore extends the EE-style story into a setting where rare-disease assumptions are not appropriate.

Methodological Contributions

- **Transmission heterogeneity:** latent infectiousness broadens variance-covariance behavior and induces shared recipient-cell shocks.
- **Learned mixing:** age-group mixing can be estimated with a flexible contact-matrix prior rather than a fixed external matrix deformation.
- **Disease-specific mechanics:** rare-disease and outbreak-driven specifications share a framework while differing in depletion, prevalence, link, and likelihood choices.

Limitations and Future Work

- Bayesian flexibility makes the model fit possible, but weakly-identified hierarchical parameters depend on workable priors.
- Simulation stress tests show sensitivity near parameter-space boundaries, especially when latent and observation-level dispersion are difficult to distinguish.
- Real-world interpretation still assumes reported incidence is an adequate proxy for true incidence.
- Future work should target prior sensitivity, faster computation for the latent layer, explicit reporting mechanisms, and richer prevalence/depletion functions.

Reporting-process extensions are a natural connection to underreporting work for EE models [[BH21](#)].

Main Empirical Lesson

- **Bottom line:** fine-grain incidence data can support inference on mixing and transmission heterogeneity when regularization, disease-specific assumptions, and latent approximations are explicit.
- Simulation: predictive gains appear when transmission heterogeneity is truly present.
- Norovirus: known assortative mixing conclusions are recovered with looser constraints and new evidence for latent heterogeneity.
- COVID-19: a bounded outbreak model identifies strong age-assortative mixing and short infectious memory in a new doubly-stratified dataset.

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Appendix: Rare-Disease Likelihood

- The rare-disease instance uses

$$\log(\delta_{tgi}) = \log\left(\frac{E_{gi}}{E_{..}}\right) + \beta_0 + \beta_g^{(G)} + \beta_i^{(I)} + \beta_i^{(S)} \sin \omega t + \beta_i^{(C)} \cos \omega t + \beta^{(\text{xmas})} x_t$$

and

$$\log(\varphi_{gi}) = \eta^{(\text{pop})} \log\left(\frac{E_{gi}}{E_{..}}\right) + \eta_0 + \eta_g^{(G)} + \eta_i^{(I)}.$$

- With $R_{tgi} \equiv 1$ and $g(\hat{X}_{tgi}, \lambda_{tgi}) = \lambda_{tgi}$,

$$\begin{aligned} \mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\eta}, \mathbf{C}, \rho, \psi, \theta) &= \prod_t \prod_{g,i} p_{\text{NB}}(Y_{tgi}^* \mid \mathbf{r}_t, \boldsymbol{\beta}, \boldsymbol{\eta}, \mathbf{C}, \rho, \psi) \\ &\quad \times p_{\text{Gamma}}(r_{tgi} \mid Y_{(t-1)gi}^*, \theta). \end{aligned}$$

Appendix: Simulation Coverage Tables

Coverage for ψ

ψ	$\theta = 0.05$		$\theta = 5$		$\theta = 15$		$\theta = 40$	
	MH	F	MH	F	MH	F	MH	F
0.05	0.81	0.92	0.00	0.72	0.00	0.53	0.00	0.66
0.50	0.90	0.84	0.09	0.88	0.00	0.85	0.00	0.86
1.00	0.90	0.86	0.27	0.88	0.00	0.86	0.00	0.85
3.00	0.87	0.85	0.65	0.89	0.14	0.87	0.00	0.84

Coverage for θ

ψ	0.05	5	15	40
0.05	0.02	0.93	0.50	0.02
0.50	0.01	0.94	0.61	0.09
1.00	0.00	0.91	0.72	0.28
3.00	0.00	0.94	0.88	0.74

Appendix: Model-fitting workloads

- Simulation fits used NUTS in Stan with four chains, 1,500 warmup iterations, and 1,500 retained draws per chain.
- Berlin norovirus fits used 16 chains, 2,000 warmup iterations, and 2,000 retained draws per chain.
- Michigan COVID-19 used the same 16-chain workflow as the norovirus analysis.
- Computation was performed on the Oregon State University College of Engineering high-performance computing cluster.

Implementation used Stan and CmdStanR [[Sta25](#); [Gab+25](#)].

Appendix: Outbreak geographic mixing

- For Michigan COVID-19, geographic weights are normalized versions of

$$w_{gg'}^{(G)} \propto \tau_g D_{gg'}^{-\rho_{g'}}.$$

- $\tau_g > 0$ represents destination attractiveness.
- $\rho_{g'} > 0$ represents source-specific distance decay and is estimated as a random effect.
- $D_{gg'}$ approximates average human distance between PUMAs using census-tract centroids and Haversine distance.